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1| theory Sqrt2
2| imports Main
3| begin
4| theorem sqrt2_not_rational:
5|   "sqrt (real 2)  $\notin$   $\mathbb{Q}$ "
6| proof
7|   let ?x = "sqrt (real 2)"
8|   assume "?x  $\in$   $\mathbb{Q}$ "
9|   then obtain m n :: nat where
10|     sqrt_rat: "?x = real m / real n" and lowest_terms: "coprime m n"
11|     by (rule Rats_abs_nat_div_natE)
12|   hence "real (m^2) = ?x^2 * real (n^2)" by (auto simp add: power2_eq_square)
13|   hence eq: "m^2 = 2 * n^2" using of_nat_eq_iff power2_eq_square by fastforce
14|   hence "2 dvd m^2" by simp
15|   hence "2 dvd m" by simp
16|   have "2 dvd n" proof
17|     from 2 dvd m obtain k where "m = 2 * k"
18|     with eq have "2 * n^2 = 2^2 * k^2" by simp
19|     hence "2 dvd n^2" by simp
20|     thus "2 dvd n" by simp
21|   qed
22|   with 2 dvd m have "2 dvd gcd m n" by (rule gcd_greatest)
23|   with lowest_terms have "2 dvd 1" by simp
24|   thus False using odd_one by blast
25| qed
26| end
27|
28|

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